FreMamba: A Frequency-Domain Mamba Model for Hyperspectral Image Change Detection

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***Abstract*—The task of hyperspectral image change detection (HSI-CD) is to identify the subtle category changes of land surfaces by utilizing the rich spectral information of bi-temporal hyperspectral images (HSIs). Recent advanced deep learning methods have improved the performance of HSI-CD. However, HSIs have the mixed-pixel phenomenon, which affects the ability of HSI-CD models to discriminate land-cover changes. In this article, a frequency-domain Mamba (FreMamba) model is proposed to precisely capture the details of the spatial and spectral changes in bi-temporal HSIs, with consideration of the above-mentioned problem. The proposed FreMamba model utilizes the capability of the Mamba model to adaptively filter the critical change information in long-range feature sequences. This is combined with frequency-domain information to enhance the low-frequency global information and high-frequency spatial detail representation, based on a dual-branch network structure. A low-frequency spectral-guided attention (LSGA) module is proposed for the low-frequency Mamba branch, where it is embedded within the Mamba block via residual connections. Global low-frequency features are aggregated by the LSGA module with noise suppression, while the spectral change discriminability of ground objects is adaptively enhanced via the subsequent Mamba blocks. In the high-frequency Mamba branch, a high-frequency spatial geometric enhancement (HSGE) module is proposed that preserves edge details by spatial frequency-domain decomposition to extract high-frequency geometric features. Experiments on three public HSI-CD datasets demonstrate that the proposed FreMamba model can accurately capture detailed change information and can outperform the existing state-of-the-art HSI-CD methods.**

***Index Terms*—Change detection, deep neural network, frequency-domain information, hyperspectral images (HSIs), Mamba.**

I. Introduction

HYPERSPECTRAL images (HSIs), owing to their broad spectral coverage and rich spectral information [1], [2], enable multitemporal HSI change detection (HSI-CD) techniques to accurately identify the subtle category changes of land surfaces [3], [4], [5]. This capability has led to their widespread application in land-use/land-cover change monitoring [6], disaster response and damage assessment [7], and urban dynamics tracking and management [8]. The traditional HSI-CD methods typically generate change prediction maps through algebra-based, transformation-based, and classification-based approaches [4], [9], [10], [11], [12].

Deep learning methods possess automatic feature extraction and high-dimensional data processing capabilities and have been extensively researched in the HSI-CD field [13], [14]. The current deep learning-based HSI-CD methods can be broadly divided into unsupervised and supervised learning paradigms. Unsupervised deep learning approaches reduce the reliance on labeled data while leveraging unlabeled multitemporal HSIs to extract spectral change features [15], [16]. These methods primarily follow two strategies. One strategy is to generate high-confidence pseudo-labels as supervisory labels [17], [18], and the other is to use a contrastive learning framework that establishes feature similarity constraints to train the change detection network [19], [20].

Supervised deep learning methods for HSI-CD leverage high-quality labeled samples to train the HSI-CD network, enabling precise extraction of discriminative spectral change features. These approaches typically employ a deep neural network to learn the hierarchical representations that capture the subtle spectral changes between multitemporal HSIs. While requiring carefully annotated ground-truth (GT) data, the supervised methods can achieve a superior performance in complex scenarios, as evidenced by the recent advanced band selection-based methods [21], [22], [23], convolutional neural networks (CNNs) [24], [25], Transformer models [26], [27], and graph neural networks (GNNs) [28], [29].

Several researchers have observed significant spectral redundancy in bi-temporal HSIs, prompting the development of band selection-based HSI-CD methods that specifically select the change-informative spectral bands [21], [22], [23]. Yao et al. [21] reduced the differences in band correlation through clustering and assigned weights to each clustered band, thereby eliminating the redundant bands. Wu et al. [22] proposed an attention-based differential band reweighting subnetwork (DBRN) that analyzes the spatial–spectral discrepancies in bi-temporal HSIs to extract discriminative features for the subsequent HSI-CD network, with the parameter optimization guided by the network. Lv et al. [23] proposed a recursive band selection algorithm that identifies sensitive spectral bands with subsequent probability density analysis of pixel pairs,

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which enhances the change detection robustness in complex scenes.

The CNN-based HSI-CD methods typically employ stacked convolutional layers to extract deep spectral change features [30], [31]. Qu et al. [30] proposed a multiscale convolutional sparse coding (MSCSC) model that extracts discriminative multiscale spectral change characteristics. Feng et al. [31] introduced a detail enhancement module to reconstruct the original HSI patches, thereby improving the spatial resolution of HSIs. This enhancement provides the subsequent change detection network with richer, detailed spatial information and textural features. Wen et al. [32] incorporated a 3-D generative adversarial network (GAN) into the HSI-CD network to enhance the discriminative ability for potential high-level change features. Zhang et al. [33] proposed a generative cross-domain change detection (GCCD) network, a GAN-based network for HSI-CD, which employs adversarial learning to mitigate domain shift interference, thereby enhancing change detection accuracy. Lu et al. [34] proposed a dual-branch CNN network to process bi-temporal HSIs through subtraction and concatenation operations separately based on a spatial attention-based symmetric feature fusion module (SFFM) to enhance the change feature representation. Jiang et al. [35] proposed a center-focused attention (CFA) module that enhances the most relevant spatial regions and spectral bands for HSI-CD. Qin et al. [36] introduced a domain alignment loss that optimizes the mean and variance matching between bi-temporal HSIs, thereby resolving the feature distribution discrepancies in HSI-CD.

The Transformer-based HSI-CD methods model long-range spectral dependencies through self-attention mechanisms to extract change information [37]. Wang et al. [38] proposed a cross-attention Transformer architecture to extract discriminative change information from bi-temporal HSIs. Dong et al. [39] utilized a Transformer model to constantly compare the cross correlation between the abundance matrices of the HSIs, thereby enhancing the extraction of difference features. Zhang et al. [40] proposed a cascaded Transformer model, which separately extracts the spatial and spectral dimension change features. In addition, a cosine similarity-based spectral constraint module was proposed to suppress the spectral distortion in the process of extracting spatial change features. To overcome the inherent limitations of patch-based learning in capturing global image context, Wang et al. [41] proposed a triplet Transformer network that establishes long-range dependencies with the other samples within the training batch, thereby extending the change contextual awareness beyond local patches. Chen et al. [42] implemented a distance-weighted spatial attention mechanism that prioritizes central pixels to suppress spectral and spatial noise and increase the discrimination between spectra with spatial change features. Li et al. [43] proposed a cross-layer adaptive fusion (CAF) strategy to enhance the change interaction between different self-attention layers. Wu et al. [44] employed the maximum discrete entropy theorem to extract bi-temporal HSI difference information and highlight the regions that exhibit significant spectral variations. Furthermore, CNNs can be incorporated to jointly extract both global and local change features, enabling comprehensive characterization of multiscale change patterns [25], [45]. In [45], a Siamese network was proposed, where the multiscale features hierarchically extracted by the CNN are progressively fused through a Transformer model to jointly capture fine-grained spatial details and high-level semantic representations. Hu et al. [25] proposed a global axial splitting (GAS) strategy that decomposes the self-attention mechanism computation along either the row-wise or column-wise spatial dimensions of the HSI, enabling computationally efficient global feature extraction while maintaining the global modeling capacity.

The GNN-based HSI-CD methods typically construct the graph representation from super-pixel segmentation of bi-temporal HSIs, followed by graph structure refinement for the discriminative change feature learning [46]. Katiyar and Manian [47] developed a dual-branch architecture integrating a super-pixel-level graph convolutional network (GCN) with a pixel-level CNN for change detection in complex scenarios. Zhao et al. [29] integrated a Transformer attention mechanism with graph node representation, achieving simultaneous learning of both long-range dependencies and local topological structures for HSI-CD.

Although the recent deep learning-based HSI-CD methods have achieved promising performances, the mixed-pixel phenomenon [48] in HSIs hinders the model's ability to further preserve discriminative spectral change features and spatial detail change features. Furthermore, the high spectral dimensionality of HSIs imposes a high computational cost when modeling the global information in HSI-CD data.

Considering the above-mentioned challenges, the proposed approach uses Mamba [49], [50], [51] as the basic model for the HSI-CD task and combines it with frequency-domain analysis to mitigate the mixed-pixel phenomenon while reducing the computational cost. The major contributions of this work are as follows.

1) A frequency-domain Mamba (FreMamba) model is proposed for HSI-CD, where the selective state-space model (SSM) is improved for the frequency space, and a dual-branch network structure is designed to extract both high-frequency and low-frequency components adaptively based on the frequency SSM, to address the mixed-pixel problem.
2) For the low-frequency information extraction, a low-frequency Mamba block is designed, which incorporates the proposed low-frequency spectral-guided attention (LSGA) module. The LSGA module condenses the low-frequency features from each feature channel while integrating spectral attention to simultaneously suppress spectral noise and mitigate the mixed-pixel problem. These enhanced features are then fused with the spectral features and fed into the SSM to adaptively enhance the discriminative spectral change features.
3) For the high-frequency information extraction, a high-frequency Mamba block is specifically designed, which consists of the high-frequency spatial geometric enhancement (HSGE) module, with the SSM to adaptively select the key spatial change information from the spatial frequency-domain decomposition.

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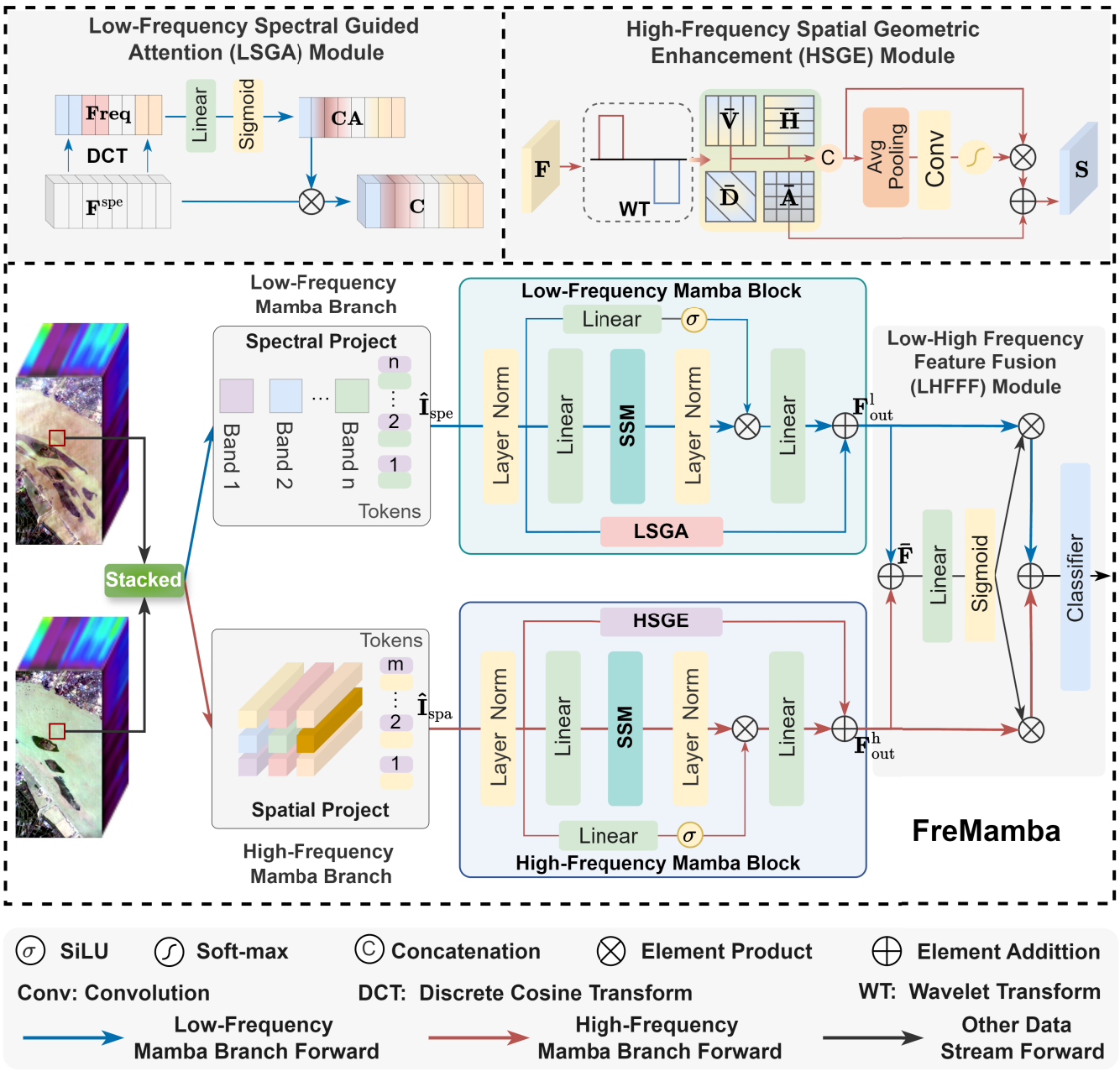


Fig. 1. Overview of the FreMamba model framework.

The rest of this article is organized as follows. In Section II, we introduce the proposed method. Section III introduces the datasets and experimental settings. An analysis is provided in Section IV. Finally, the conclusion of this study is provided in Section V.

II. Proposed Method

Fig. 1 displays the flowchart of the proposed FreMamba model, which is a dual-branch architecture. Given bi-temporal HSIs, the image patches $\mathbf{X_1}$,$\mathbf{X_2}$ for each time phase are, respectively, cropped from the same position of the original HSIs. $\mathbf{X}_1$ and $\mathbf{X}_2$ are then stacked to obtain $\mathbf{X} \in {}^{2C\times H\times W}$, where H and W are the height and width of the patch, and C is the number of bands of each HSI. Each branch then takes $\mathbf{X}$ as the input. The low-frequency Mamba branch consists of N low-frequency Mamba blocks. Each block incorporates an SSM and LSGA module designed to extract low-frequency global spectral features and mitigate spectral noise. Subsequently, the residual connections enable the SSM to adaptively refine the selection of the critical low-frequency features from the LSGA module. The high-frequency Mamba branch consists of N high-frequency Mamba blocks. Each block employs the SSM and HSGE modules to capture the high-frequency local spatial features, thereby enhancing the model's capability to preserve the geometric change details of surface objects. Finally, a lightweight attention mechanism is introduced that facilitates feature interaction between the low-frequency and high-frequency Mamba branches. The integrated features are then fed into a fully connected layer to obtain patch-wise change detection results.

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A. Low-Frequency Mamba Branch of FreMamba

HSIs suffer from the mixed-pixel problem due to their inherent limitation in spatial resolution. This issue exacerbates the challenges of spectral distinguishability and pseudo-changes in HSI-CD. To address these issues, the LSGA module is proposed in the low-frequency Mamba branch. This module employs the discrete cosine transform (DCT) to condense the low-frequency features of each feature channel, thereby effectively suppressing the spectral noise [52]. Subsequently, spectral attention is applied to enhance the discriminative capability for true changes. The fused features from both the LSGA and SSM outputs are then fed into the next low-frequency Mamba block to enable adaptive refinement of the critical low-frequency feature selection.

First, the input is $\mathbf{X}$ reshaped into $HW \times 2C$, and a linear projection is performed based on a convolution operation

$$\mathbf{I}_{\text{spe}} = \text{Conv}\left[\text{Reshape}\left(\mathbf{X}\right)\right] \tag{1}$$

where Reshape(•) is the reshape operation and Conv(•) is the 1-D convolution operation, in which the kernel size of the convolution operation is set to the spectral patch size P_{spe}. The input data $\mathbf{X}$ are partitioned into $N_{\text{spe}} = 2C/P_{\text{spe}}$ tokens in the spectral dimension. After this, the learnable classification token $\mathbf{I}_{\text{cls}}$ followed with a 1-D positional embedding $\mathbf{E}_{\text{pos}}$ is embedded into the sequence

$$\hat{\mathbf{I}}_{\text{spe}} = \text{Concat}\left(\mathbf{I}_{\text{spe}}, \mathbf{I}_{\text{cls}}\right) + \mathbf{E}_{\text{pos}} \tag{2}$$

$$\mathbf{F}^{\text{spe}} = \text{LN}\left(\hat{\mathbf{I}}_{\text{spe}}\right) \tag{3}$$

where Concat(•) is the concatenation operation, LN(•) denotes the layer normalization operation, and $\mathbf{F}^{\text{spe}}$ represents the input feature sent into the low-frequency Mamba block.

For each feature channel in $\mathbf{F}^{\text{spe}}$, the corresponding 2D-DCT frequency component is assigned as follows:

$$B_{h,w}^{i,j} = \cos\left(\frac{\pi u}{H_F}\left(i+\frac{1}{2}\right)\right)\cos\left(\frac{\pi v}{H_F}\left(j+\frac{1}{2}\right)\right)$$
$$i \in \{0,1,\ldots,H_F-1\},\ \ j \in \{0,1,\ldots,W_F-1\} \tag{4}$$

where H_F and W_F are the height and width of $\mathbf{F}^{\text{spe}}$, respectively. u and v represent the predefined frequency combinations assigned to the different spectral channels. The 2D-DCT frequency feature of each channel is then obtained

$$\text{Freq}^i = 2\text{D} - \text{DCT}\left(\mathbf{F}^{\text{spe}}\right) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} F_{h,w}^{\text{spe}} B_{h,w}^{i,j}. \tag{5}$$

All the frequency features in the LSGA module can then be obtained by a concatenation operation

$$\mathbf{Freq} = \text{Concat}\left(\text{Freq}^0, \text{Freq}^1, \ldots, \text{Freq}^{n-1}\right). \tag{6}$$

The linear layer and sigmoid function are then used to convert the frequency spectrum into a spectral attention weight $\mathbf{CA}$

$$\mathbf{CA} = \text{Sigmoid}\left[Fc\left(\mathbf{Freq}\right)\right] \tag{7}$$

where $Fc(\bullet)$ represents the linear layer. The attention weight $\mathbf{CA}$ are multiplied with the input feature $\mathbf{F}^{\text{spe}}$ of the LSGA module to generate the attention feature $\mathbf{C}$

$$\mathbf{C} = \mathbf{F}^{\text{spe}} \cdot \mathbf{CA}. \tag{8}$$

The output feature $\mathbf{C}$ of the LSGA module and output feature $\mathbf{F}_{\text{ssm}}^{l}$ of the SSM are then fused through element-wise addition, forming the final output feature $\mathbf{F}_{\text{out}}^{l}$ of the low-frequency Mamba branch

$$\mathbf{F}_{\text{out}}^{l} = \mathbf{C} + \mathbf{F}_{\text{ssm}}^{l}. \tag{9}$$

B. High-Frequency Mamba Branch of FreMamba

Due to the limited spatial resolution of HSIs, a high-frequency Mamba branch combining the HSGE module with the SSM is proposed to adaptively discriminate the true change details from the pseudo-changes, from the aspect of the spatial domain.

The HSGE module extracts the high-frequency spatial features through spatial frequency-domain decomposition. These features capture the rich geometric texture information of changing targets, enhancing the model's sensitivity to detailed spatial patterns. Subsequently, a spatial attention mechanism is incorporated to emphasize the salient regions of interest. The processed high-frequency features are then fused with the low-frequency components, maintaining global contextual awareness while effectively suppressing the pseudo-changes caused by geometric distortion.

For the input patch $\mathbf{X}$, a 2-D convolution operation is used to project and flatten it along the spatial dimension and obtain the spatial input tokens $\hat{\mathbf{I}}_{\text{spa}}$. After layer normalization of $\hat{\mathbf{I}}_{\text{spa}}$, the spatial input $\mathbf{F}^{\text{spa}}$ is acquired.

For each feature channel in $\mathbf{F}^{\text{spa}}$, the spatial frequency-domain decomposition operation is calculated as follows:

$$\mathbf{D}_b\left(i,j\right) = \mathbf{F}^{\text{spa}}\left(2i-1,j\right) - \mathbf{F}^{\text{spa}}\left(2i,j\right) \tag{10}$$

$$\mathbf{A}_b\left(i,j\right) = \frac{1}{2}\left[\mathbf{F}^{\text{spa}}\left(i,2j-1\right) + \mathbf{F}^{\text{spa}}\left(i,2j\right)\right] \tag{11}$$

where $\mathbf{D}_b(i,j)$ and $\mathbf{A}_b(i,j)$ are the high-frequency component and low-frequency component of feature channel b in the ith row and jth column, respectively. The Haar wavelet transform (WT) [53], [54] is then applied to multiscale decompose the features into three high-frequency features and one low-frequency feature

$$\bar{\mathbf{A}} = \frac{1}{2}\left[\mathbf{A}_b\left(i,2j-1\right) + \mathbf{A}_b\left(i,2j\right)\right] \tag{12}$$

$$\bar{\mathbf{H}} = \frac{1}{2}\left[\mathbf{D}_b\left(i,2j-1\right) - \mathbf{D}_b\left(i,2j\right)\right] \tag{13}$$

$$\bar{\mathbf{V}} = \mathbf{A}_b\left(i,2j-1\right) - \mathbf{A}_b\left(i,2j-1\right) \tag{14}$$

$$\bar{\mathbf{D}} = \mathbf{D}_b\left(i,2j-1\right) - \mathbf{D}_b\left(i,2j\right) \tag{15}$$

where $\bar{\mathbf{A}}$ is the low-frequency feature, $\bar{\mathbf{H}}$ is the horizontal high-frequency feature, $\bar{\mathbf{V}}$ is the vertical high-frequency feature, and $\bar{\mathbf{D}}$ is the diagonal high-frequency feature.

A spatial attention mechanism then follows to better focus on the geometric texture information. The high-frequency features from the horizontal, vertical, and diagonal orientations

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TABLE I
DETAILS OF THE THREE PUBLIC DATASETS

	River	Farmland	Hermiston
Size	463×241	450×140	390×200
Channels	198	155	242
Changed	9698	18,277	19,972
Unchanged	101,885	44,723	58,028
Data of T1	May 3, 2013	May 3, 2006	May 1, 2004
Data of T2	Dec 31, 2013	Apr 23, 2007	May 8, 2007
Location	Jiangsu province, China	Jiangsu province, China	Umatilla county, USA

are then concatenated to form a comprehensive high-frequency feature $\mathbf{F}_H$

$$\mathbf{F}_H = \text{Concat}\left(\bar{\mathbf{H}}, \bar{\mathbf{V}}, \bar{\mathbf{D}}\right). \tag{16}$$

An average pooling operation is then applied to transform the feature into the attention weight

$$\mathbf{SA}_{\text{avg}} = \text{AvgPool}\left(\mathbf{F}_H\right) \tag{17}$$

where AvgPool(•) represents the average pooling operation. Convolution and Softmax operations are then used to convert the pooling feature into a spatial attention weight $\mathbf{SA}$

$$\mathbf{SA} = \text{Softmax}\left[\text{Conv}\left(\mathbf{SA}_{\text{avg}}\right)\right]. \tag{18}$$

The high-frequency feature $\mathbf{F}_H$ is enhanced through multiplication with the attention weight $\mathbf{SA}$. This refined feature is then fused with the low-frequency feature $\bar{\mathbf{A}}$ by element-wise addition, thereby preserving the global contextual information

$$\mathbf{S} = \mathbf{F}_H \cdot \mathbf{SA} + \bar{\mathbf{A}}. \tag{19}$$

The output feature $\mathbf{S}$ of the HSGE module and output feature $\mathbf{F}^h_{\text{ssm}}$ of the SSM are then fused through element-wise addition, forming the final output feature $\mathbf{F}^h_{\text{out}}$ of the high-frequency Mamba branch

$$\mathbf{F}^h_{\text{out}} = \mathbf{S} + \mathbf{F}^h_{\text{ssm}}. \tag{20}$$

C. LHFFF and Change Detection

A low-high frequency feature fusion (LHFFF) module based on a lightweight attention mechanism is proposed to adaptively select and enhance both the high-frequency and low-frequency change features. First, the aggregated features obtained from both branches are summed through weighted summation

$$\bar{\mathbf{F}} = \frac{1}{2}\left(\mathbf{F}^l_{\text{out}} + \mathbf{F}^h_{\text{out}}\right). \tag{21}$$

The aggregated features are then converted into the attention weight by the linear layer and the sigmoid function, and the output feature is acquired as follows:

$$\mathbf{F}_{\text{CA}} = \text{Sigmoid}\left[Fc\left(\bar{\mathbf{F}}\right)\right] \tag{22}$$

$$\hat{\mathbf{F}}^l_{\text{out}} = \mathbf{F}_{\text{CA}} \cdot \mathbf{F}^l_{\text{out}} \tag{23}$$

$$\hat{\mathbf{F}}^h_{\text{out}} = \mathbf{F}_{\text{CA}} \cdot \mathbf{F}^h_{\text{out}}. \tag{24}$$

The fusion feature is input into the classification layer to obtain the change detection prediction $\mathbf{Y}$

$$\mathbf{Y} = \text{Softmax}\left[Fc\left(\hat{\mathbf{F}}^l_{\text{out}} + \hat{\mathbf{F}}^h_{\text{out}}\right)\right]. \tag{25}$$

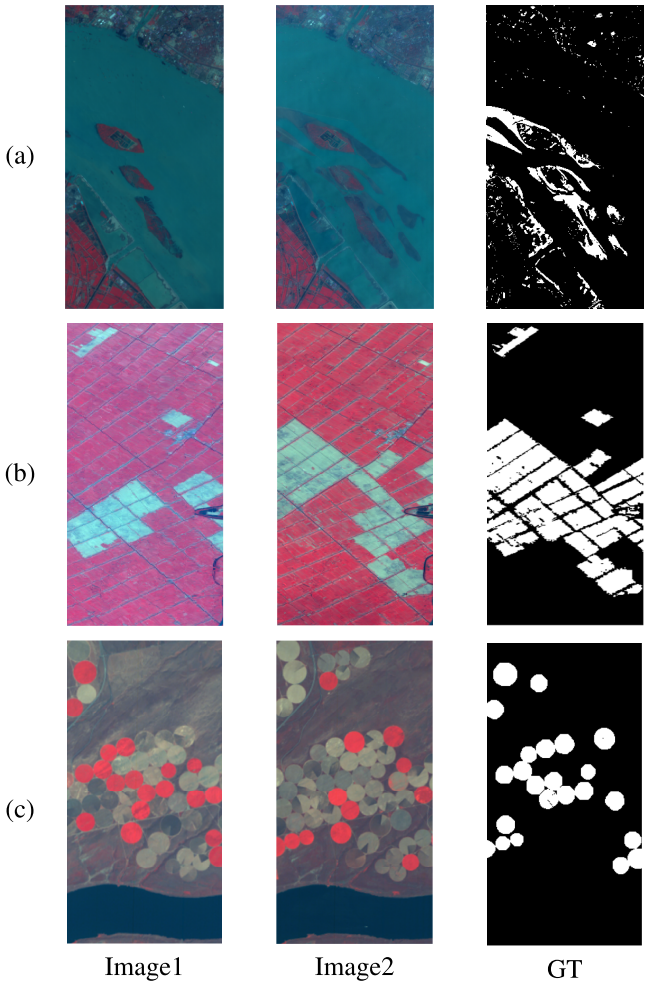


Fig. 2. Pseudo-color and GT images of the three HSI-CD datasets. (a) River dataset. (b) Farmland dataset. (c) Hermiston dataset. For each row, from left to right, are the HSI acquired at time 1, the HSI acquired at time 2, and the GT labels.

III. EXPERIMENTAL SETTINGS

A. Descriptions of the Datasets

To validate the effectiveness of the proposed FreMamba model, three public HSI-CD datasets were used for the testing: River [55], Farmland [56], and Hermiston.[1] These

[1]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

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TABLE II

QUANTITATIVE RESULTS OF THE ABLATION STUDY RELATED TO THE PROPOSED MODULES ON THE RIVER, FARMLAND, AND HERMISTON DATASETS, WITH THE HIGHEST SCORES HIGHLIGHTED IN BOLD

Dataset	Structure	Model	LSGA	HSGE	LHFFF	OA (%)	Kappa (%)	Pr (%)	R (%)	F1 (%)
River	Single branch	Low-frequency Mamba branch				97.47	83.34	**89.07**	80.78	84.72
			✓			97.61	84.51	88.39	83.38	85.81
	Single branch	High-frequency Mamba branch				97.54	84.02	86.35	84.29	85.31
				✓		97.59	84.47	87.12	84.34	85.71
	Dual branch	FreMamba				97.56	84.13	86.54	84.54	85.53
			✓			97.62	84.67	88.41	83.58	85.93
				✓		97.59	84.83	87.34	84.93	86.12
					✓	97.55	84.33	87.03	84.60	85.80
			✓	✓		97.58	84.97	87.31	85.17	86.23
			✓		✓	97.61	84.75	87.26	84.89	86.06
				✓	✓	97.65	85.08	87.47	85.28	86.36
			✓	✓	✓	**97.71**	**85.42**	87.76	**85.60**	**86.67**
Farmland	Single branch	Low-frequency Mamba branch				97.58	94.60	94.81	96.95	95.87
			✓			98.09	95.37	95.77	97.65	96.70
	Single branch	High-frequency Mamba branch				97.83	94.80	94.78	**97.93**	96.33
				✓		98.04	95.24	96.49	96.75	96.62
	Dual branch	FreMamba				97.71	94.46	95.69	96.46	96.08
			✓			98.13	95.45	96.51	96.73	96.62
				✓		97.92	95.23	95.89	96.83	96.36
					✓	98.13	95.39	96.52	96.68	96.60
			✓	✓		98.11	95.47	96.61	96.93	96.77
			✓		✓	98.09	95.49	96.59	96.85	96.72
				✓	✓	98.14	95.28	96.43	97.19	96.81
			✓	✓	✓	**98.20**	**95.63**	**96.77**	97.03	**96.90**
Hermiston	Single branch	Low-frequency Mamba branch				98.46	95.08	95.16	95.52	95.34
			✓			98.64	95.99	96.36	95.82	96.09
	Single branch	High-frequency Mamba branch				98.50	95.84	96.62	95.70	96.16
				✓		98.70	96.03	96.71	95.87	96.29
	Dual branch	FreMamba				99.02	95.63	96.26	96.12	96.19
			✓			99.10	96.00	96.28	96.75	96.51
				✓		99.09	95.92	97.00	95.88	96.44
					✓	98.73	96.17	96.59	96.23	96.41
			✓	✓		99.17	96.27	**96.88**	96.64	96.76
			✓		✓	99.03	96.29	96.63	96.85	96.74
				✓	✓	99.12	96.41	96.73	96.97	96.85
			✓	✓	✓	**99.21**	**96.51**	96.78	**97.14**	**96.96**

datasets were all obtained by the Hyperion sensor on the Earth Observing-1 (EO-1) satellite [57]. The Hyperion sensor provides a total of 242 spectral bands, with a spectral range from 0.4 to 2.5 μm. The spatial resolution of this data is 10 m, and the spectral resolution is 30 nm [58]. The details of the three HSI-CD datasets are given in Table I. Fig. 2 shows the images of the three-channel pseudo-color composites (bands 39, 19, and 9 as RGB [41]) and the GT labels.

B. Implementation Details

The hyperparameter settings and the training details for the proposed FreMamba model used in the experiments are provided in this part.

1) Network Structure and Parameters of FreMamba: In the low-frequency and high-frequency Mamba branches, the number of Mamba blocks N for change feature extraction was set to 4, based on the parameter analysis part. Each Mamba block consists of a linear layer, layer normalization, a sigmoid linear unit (SiLU) activation function [59], and the SSM. This architecture allows for efficient feature extraction and the enhancement of the interaction between the low-frequency and high-frequency features, thereby improving the performance in capturing the changes.

2) Training Settings: The experiments described in this article were conducted on a workstation equipped with an NVIDIA GeForce 3090 GPU and implemented in the Python environment. The same parameters were used for all three datasets, where the patch size was set to 27, the batch size was set to 512, the training epochs were set to 170, and the initial learning rate was set to 0.0005. The Adam optimizer was used as the optimizer, with a weight decay of 0.0001. We randomly selected 5% of each dataset as the training samples. The experiments were repeated five times, and the average value for each evaluation index was calculated.

C. Evaluation Metrics

The overall accuracy (OA), kappa coefficient (kappa), precision (Pr), recall (R), and $F1$-score ($F1$) are used here as indicators to evaluate the model performance.

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The OA represents the proportion of correctly detected pixels and is calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{26}$$

where TP, TN, FP, and FN are the number of true positives, true negatives, false positives, and false negatives, respectively.

Kappa is a statistical measure that accounts for the interrater reliability, thereby offering a more objective assessment of the prediction results. Kappa is calculated as follows:

$$P_e = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN}) + (\mathrm{FN} + \mathrm{TN})(\mathrm{FP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} \tag{27}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - P_e}{1 - P_e}. \tag{28}$$

The $F1$ is the harmonic average of the precision (Pr) and recall (R), which can evaluate the accuracy of prediction results in a more balanced way

$$F1 = \frac{2 \times \mathrm{Pr} \times R}{\mathrm{Pr} + R} \tag{29}$$

$$\mathrm{Pr} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{30}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{31}$$

IV. Experimental Results and Analysis

A. Ablation Study and Parameter Analysis

1) Ablation Study of the Proposed Modules: The effectiveness of each Mamba branch in the proposed FreMamba model was analyzed on the three datasets. The accuracy results are presented in Table II.

For the River dataset, the first two rows of Table II correspond to the single-branch configuration of the network, which utilizes only the low-frequency Mamba branch of FreMamba and excludes the LHFFF module. This branch processes the input bi-temporal HSIs by modeling their spectral dimension. For the River dataset, the first two rows in the table represent the single low-frequency Mamba branch structure of the network, which models the input bi-temporal HSIs in the spectral dimension. The first row represents the network architecture without utilizing low-frequency features. The inclusion of the LSGA module results in a 1.17% increase in kappa and a 1.09% improvement in $F1$. The LSGA module extracts the global low-frequency features of the HSIs through DCT, which effectively suppresses the spectral noise while enhancing the model's capability to extract discriminative spectral change features. The improvement in accuracy indicates that the LSGA module enhances the network's ability to extract change information.

The third and fourth rows in Table II represent the use of the single high-frequency branch structure as the network, which focuses on extracting the spatial change information. The third and fourth rows in the table represent the single high-frequency branch structure of the network, which focuses on extracting the spatial change information. The third row represents the network architecture without utilizing high-frequency features. After incorporating the HSGE module, improvements in both kappa (0.45%) and $F1$ (0.40%) are seen. The Mamba model has powerful global modeling capabilities for long-sequence data, but lacks sufficient ability to express local, detailed change features. The HSGE module extracts high-frequency geometric information from the features using the Haar WT, while applying a spatial attention mechanism for further enhancement. This effectively compensates for the missing local spatial detail features in the Mamba block.

The last eight rows in Table II correspond to the dual-branch architecture of the model, wherein the low-frequency Mamba branch captures spectral changes, while the high-frequency Mamba branch models spatial changes. The LHFFF module is incorporated to adaptively select and enhance the change features derived from these two branches.

The dual-branch architecture demonstrates superior performance compared to both single-branch structures, indicating its enhanced capability to effectively incorporate both spectral and spatial change information from bi-temporal HSIs. Based on this architecture, the individual incorporation of the LSGA, HSGE, and LHFFF modules each contributes to performance improvement. The LSGA and HSGE modules enhance the model's ability to suppress false changes by extracting low–high frequency change features, respectively. The LHFFF module employs a lightweight attention mechanism to adaptively fuse the low-high frequency features, thereby improving the generalization capability of the model. When all three modules are incorporated together, the model achieves its best performance, yielding a kappa of 85.42% and an $F1$ of 86.67%. Compared to the baseline model, the generalization ability of the FreMamba model has been significantly improved. For the Farmland and Hermiston datasets, the performance was generally similar to the performance with the River dataset.

2) Parameter Analysis of the Number of Mamba Blocks N: The effect of different numbers of Mamba blocks N in the FreMamba model and the effect of the size of the HSI patches were analyzed on the River dataset. The experimental results are reported in Table III. When the number of Mamba blocks N is set to 4, and the patch size is set to 27, FreMamba achieves the highest accuracy. The image patch size is set to 27, which allows the model to capture richer contextual information. Moreover, owing to the selective mechanism of Mamba, the model enhances the extraction of discriminative spectral features while reducing the influence of redundant spectral bands, thereby improving the discrimination capability for land-cover change targets.

3) Parameter Analysis of the Spectral Patch Size P_{Spe} and Spatial Patch Size P_{Spa}: The influence of varying the spectral patch size P_{spe} and spatial patch size P_{spa} in FreMamba was evaluated on the River dataset. The experimental results, summarized in Fig. 3, indicate that the model achieves the best performance when P_{spa} is set to 3 and P_{spe} is set to 2. This parameter configuration is adopted in the proposed FreMamba model.

B. Comparison Results and Analysis

To validate the effectiveness of the proposed method, the FreMamba model was compared with several

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TABLE III

QUANTITATIVE RESULTS OF FREMAMBA ON THE RIVER DATASET WITH DIFFERENT NUMBERS OF MAMBA BLOCKS AND PATCH SIZES, WITH THE HIGHEST SCORES HIGHLIGHTED IN BOLD

Number of Mamba block N	Patch size	OA (%)	Kappa (%)	Pr (%)	R (%)	F1 (%)
3	23	97.24	82.36	85.51	82.27	83.86
	25	97.44	83.42	86.09	84.62	85.35
	27	97.61	84.76	87.19	84.98	86.07
	29	97.22	82.07	85.71	81.57	83.59
4	23	97.31	82.75	85.76	82.75	84.23
	25	97.42	84.23	86.54	84.25	85.38
	27	**97.71**	**85.41**	87.76	85.61	**86.67**
	29	97.30	82.68	85.96	82.41	84.15
5	23	97.36	83.11	86.24	82.92	84.55
	25	97.47	83.93	87.08	83.25	85.12
	27	97.71	85.34	**87.94**	85.28	86.59
	29	97.32	82.80	86.06	82.53	84.26
6	23	97.45	83.62	86.94	83.16	85.01
	25	97.16	84.31	86.18	84.59	85.38
	27	97.66	85.20	87.19	**85.76**	86.47
	29	97.32	82.83	85.87	82.79	84.30

TABLE IV

COMPARATIVE EXPERIMENTAL RESULTS OF THE DIFFERENT HSI-CD METHODS WITH 5% TRAINING SAMPLES ON THE RIVER DATASET, WITH THE HIGHEST SCORES HIGHLIGHTED IN BOLD

Method	OA (%)	Kappa (%)	Pr (%)	R (%)	F1 (%)
MP-ConvLSTM [25]	**98.96**	73.47	79.54	72.16	75.67
CSANet [61]	95.97	78.83	86.94	75.68	80.92
SST-Former [39]	96.96	79.58	87.49	75.79	81.22
ML-EDAN [62]	97.09	81.06	85.91	79.61	82.64
CDFormer [38]	97.33	82.33	85.87	82.56	84.18
DIEFEN [63]	97.29	81.08	**93.85**	73.63	82.52
MFCEN [46]	97.54	84.20	88.66	83.16	85.82
AIWSEN [45]	97.19	81.37	88.35	78.06	82.89
FreMamba	97.71	**85.42**	87.76	**85.59**	**86.66**

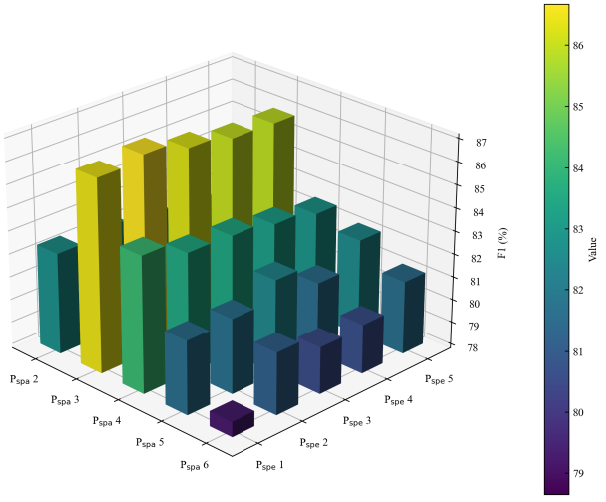


Fig. 3. Experimental results of the proposed FreMamba model on the River dataset, to evaluate the impact of different spectral patch sizes P_{spe} and spatial patch sizes P_{spa}.

state-of-the-art HSI-CD methods: MP-ConvLSTM [24], CSANet [60], SST-Former [39], ML-EDAN [61], CDFormer [37], DIEFEN [62], MFCEN [45], and AIWSEN [44]. Among these methods, MP-ConvLSTM, CSANet, ML-EDAN, and DIEFEN are CNN-based methods, while SST-Former, CDFormer, MFCEN, and AIWSEN are Transformer-based methods.

1) Result and Analysis for the River Dataset: Table IV lists the accuracies of the various methods on the River dataset. The FreMamba model achieves the highest OA, kappa, and $F1$, with values of 97.71%, 85.42%, and 86.66%, respectively, surpassing both the CNN-based and Transformer-based methods. This demonstrates the potential of the Mamba model's SSM in effectively extracting change information from bi-temporal HSIs. In comparison, MFCEN achieves a higher Pr, indicating that the proposed method misclassifies some background or negative class regions as positive. However, FreMamba has a higher R, meaning that it reduces the number of missed targets (FN) while maintaining a good level of completeness. FreMamba outperforms MFCEN in kappa and $F1$ by 1.22% and 0.84%, respectively.

The visual results for the River dataset are shown in Fig. 4. The River dataset presents a challenging task due to its small and scattered change regions, testing the model's ability to balance the long-range dependencies and local details. Compared to the prediction maps generated by the other HSI-CD methods, FreMamba exhibits significantly fewer false alarms and FPs, indicating its superior robustness in terms of both missed detection rate and false detection rate. For the appearance of the small-scale change areas in the center of

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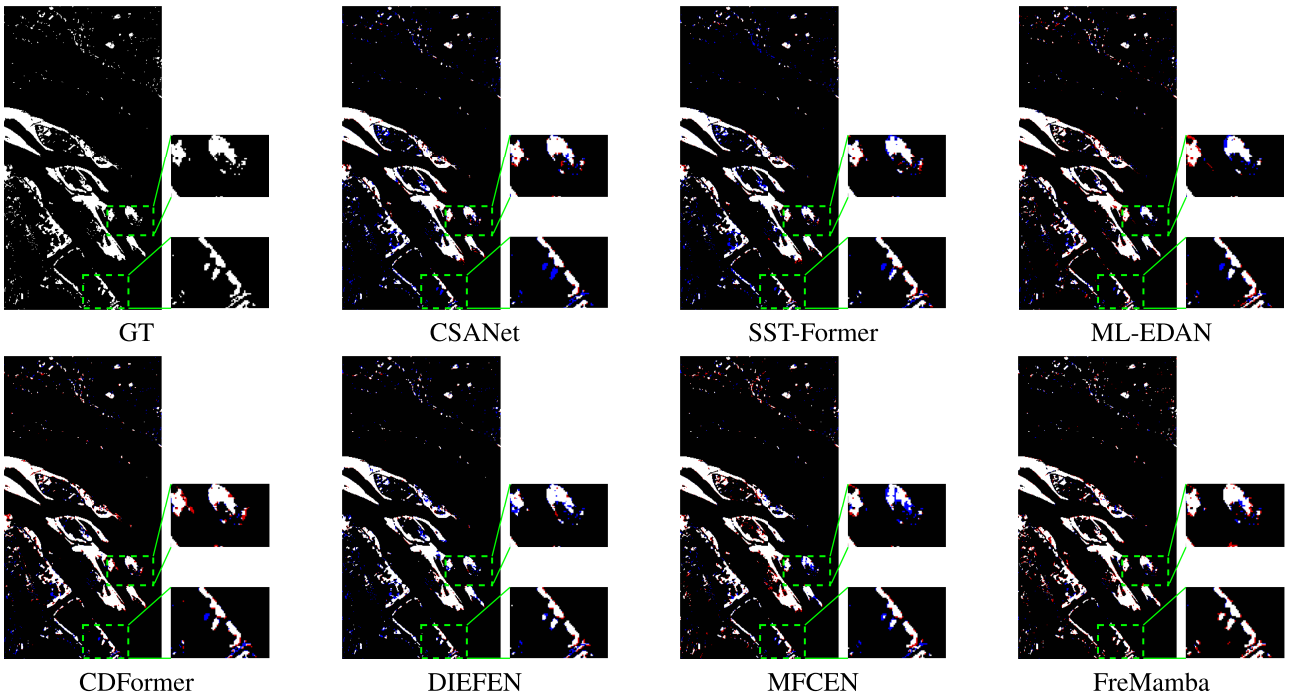


Fig. 4. HSI-CD prediction maps of all the methods on the River dataset. (Left to right) and (top to bottom) shows the ground truth, the CSANet prediction results, the SST-Former prediction results, the ML-EDAN prediction results, the CDFormer prediction results, the DIEFEN prediction results, the MFCEN prediction results, and the FreMamba prediction results. The FN and FP pixels are denoted in red and blue colors, respectively.

TABLE V

COMPARATIVE EXPERIMENTAL RESULTS OF THE DIFFERENT HSI-CD METHODS WITH 5% TRAINING SAMPLES ON THE FARMLAND DATASET, WITH THE HIGHEST SCORES HIGHLIGHTED IN BOLD

Method	OA (%)	Kappa (%)	Pr (%)	R (%)	F1 (%)
MP-ConvLSTM [25]	96.11	90.75	90.22	97.09	93.53
CSANet [61]	96.57	94.55	94.75	**97.61**	96.16
SST-Former [39]	97.78	94.65	94.99	97.48	96.22
ML-EDAN [62]	97.18	93.22	93.93	96.54	95.22
CDFormer [38]	97.62	94.23	95.43	96.21	95.82
DIEFEN [63]	97.62	94.28	94.49	97.48	95.96
MFCEN [46]	97.83	94.75	95.68	96.88	96.28
AIWSEN [45]	98.12	95.44	96.74	96.79	96.76
FreMamba	**98.20**	**95.64**	**96.77**	97.05	**96.91**

the riverway on the right side of the image, the proposed FreMamba model effectively detects the edge contours of these small-scale change areas. In addition, the subtle changes in the riverbank at the bottom of the image are also accurately detected by FreMamba. This demonstrates that the proposed FreMamba model has a clear advantage in extracting subtle change information and preserving fine boundary details. This is enabled by the HSGE module, which extracts high-frequency geometric features of the target using the Haar WT, supplementing the local geometric details not considered in the Mamba model. As a result, this enhancement improves the model's generalization ability in complex spatial scenes, making it more robust and effective in detecting fine details in varied landscapes.

2) Result and Analysis for the Farmland Dataset: The accuracy performance of the various methods on the Farmland dataset is provided in Table V. The proposed FreMamba model achieves the best accuracy, with an OA of 98.20%, kappa of 95.64%, Pr of 96.77%, and $F1$ of 96.91%. In comparison, CSANet achieves a higher R of 97.61%, whereas the proposed FreMamba model attains a superior Pr of 96.77%, demonstrating its lower false detection rate. Compared to AIWSEN, which is the second-place model, the scores of FreMamba are 0.2% higher in kappa and 0.15% higher in $F1$. This advantage can be attributed to the selective mechanism of Mamba, which efficiently extracts change information and enhances the model's ability to capture long-range dependencies. The proposed integration of the LSGA and HSGE modules effectively suppresses spectral noise while enhancing the local spatial detail preservation. This synergistic combination significantly mitigates the mixed-pixel effects, thereby improving the model's generalization capability in complex scenarios.

The visual results for the Farmland dataset are shown in Fig. 5. The Farmland dataset primarily captures agricultural changes, but its coarse farmland boundaries test the

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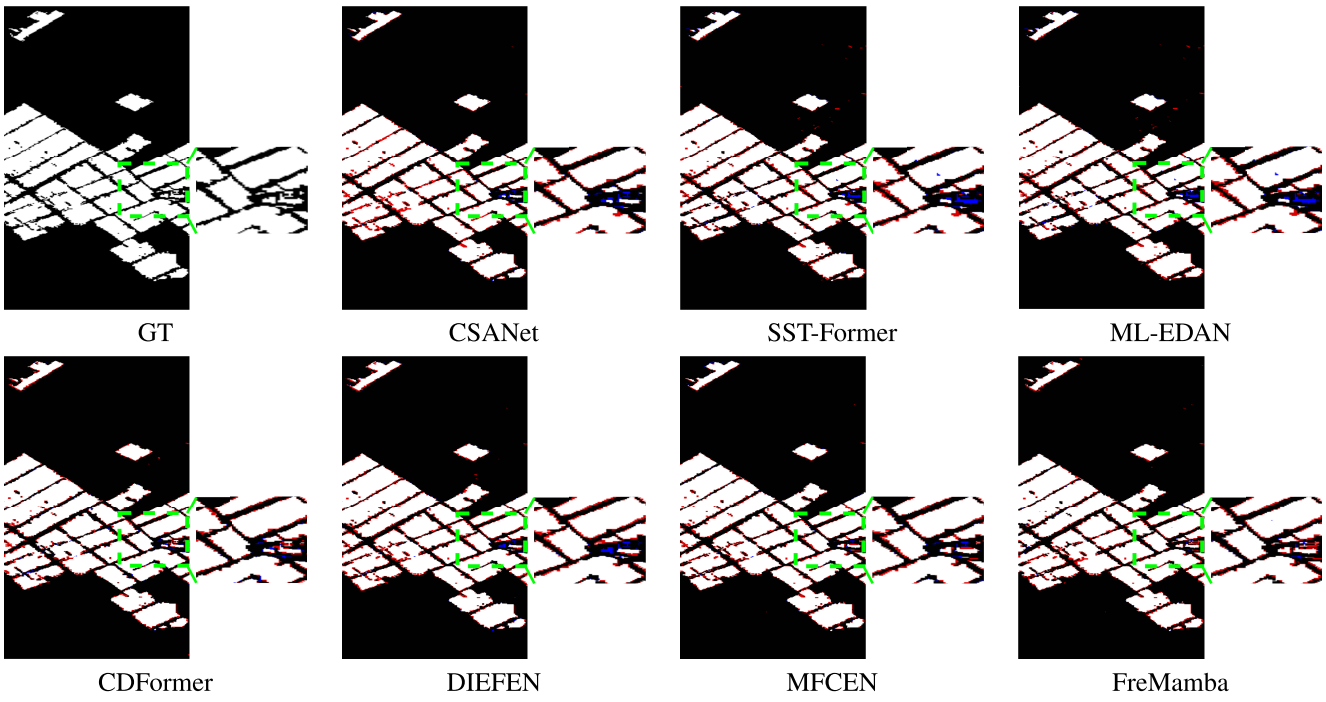


Fig. 5. HSI-CD prediction maps of all the methods on the Farmland dataset. (Left to right) and (top to bottom) shows the ground truth, the CSANet prediction results, the SST-Former prediction results, the ML-EDAN prediction results, the CDFormer prediction results, the DIEFEN prediction results, the MFCEN prediction results, and the FreMamba prediction results. The FN and FP pixels are denoted in red and blue colors, respectively.

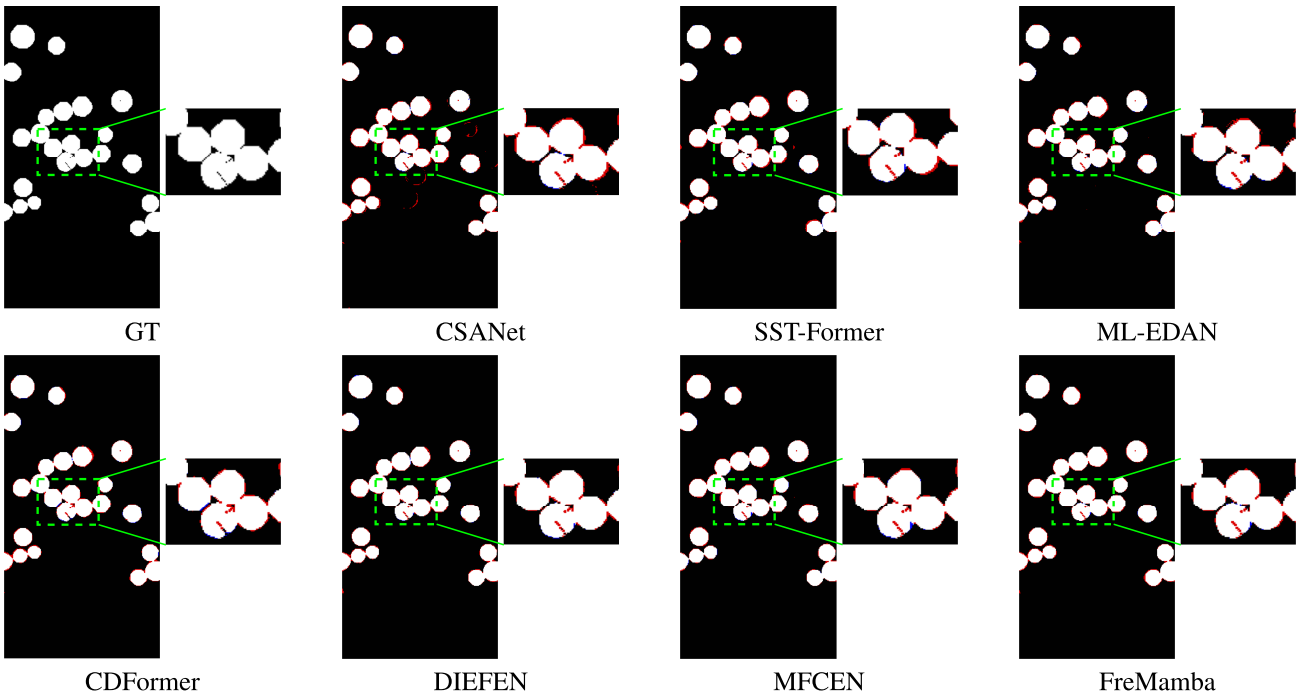


Fig. 6. HSI-CD prediction maps of all the methods on the Hermiston dataset. (Left to right) and (top to bottom) shows the ground truth, the CSANet prediction results, the SST-Former prediction results, the ML-EDAN prediction results, the CDFormer prediction results, the DIEFEN prediction results, the MFCEN prediction results, and the FreMamba prediction results. The FN and FP pixels are denoted in red and blue colors, respectively.

model's edge preservation ability. The prediction results of FreMamba for the farmland change areas show clear and sharp edges, with full internal areas, indicating that the proposed FreMamba model has an advantage in edge detail preservation. This is made possible by FreMamba's distinct modeling of both the spatial and spectral aspects of the HSIs. In addition, in the subtle and complex change region located on the left side of the image's center, FreMamba effectively detects the changes in that area. This highlights the benefit of incorporating the frequency-domain Haar

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TABLE VI
COMPARATIVE EXPERIMENTAL RESULTS OF THE DIFFERENT HSI-CD METHODS WITH 5% TRAINING SAMPLES ON THE HERMISTON DATASET, WITH THE HIGHEST SCORES HIGHLIGHTED IN BOLD

Method	OA (%)	Kappa (%)	Pr (%)	R (%)	F1 (%)
MP-ConvLSTM [25]	98.85	94.98	93.11	98.31	95.64
CSANet [61]	98.97	94.52	95.89	96.41	96.15
SST-Former [39]	98.99	95.49	95.85	96.27	96.06
ML-EDAN [62]	98.88	95.02	95.12	96.19	95.65
CDFormer [38]	98.82	96.00	95.98	95.76	95.87
DIEFEN [63]	99.18	96.37	96.72	96.94	96.83
MFCEN [46]	99.16	96.27	96.33	**97.17**	96.75
AIWSEN [45]	98.93	95.26	95.11	96.64	95.87
FreMamba	**99.22**	**96.51**	**96.78**	97.14	**96.96**

TABLE VII
TOTAL PARAMETERS AND FLOPS OF THE DIFFERENT METHODS

Method	Parameters	FLOPs
MP-ConvLSTM [25]	4.57M	6.51M
CSANet [61]	2.45M	1.15G
SST-Former [39]	2.54M	156.68M
ML-EDAN [62]	88.52M	854.75M
CDFormer [38]	307.97K	404.01M
DIEFEN [63]	77.41M	4.34G
MFCEN [46]	8.21M	139.09M
AIWSEN [45]	25.41M	1.09G
Low-frequency Mamba branch	136.13K	47.72M
High-frequency Mamba branch	214.79K	44.42M
FreMamba	492.68K	92.29M

WT, which enhances the model's ability to preserve subtle changes.

3) Result and Analysis for the Hermiston Dataset: The accuracy performance of the different methods on the Hermiston dataset is presented in Table VI. Similar to the Farmland dataset scenario, the proposed FreMamba model also achieves the best performance for this dataset, with an OA of 99.22%, kappa of 96.51%, Pr of 96.78%, and $F1$ of 96.96%. Similarly, compared to both the CNN-based and Transformer-based methods, FreMamba shows performance improvements in all the evaluation metrics. These improvements demonstrate that the Mamba model possesses exceptional capabilities in extracting the HSI change features.

Fig. 6 illustrates the detection results of the various methods on the Hermiston dataset. The changes in the Hermiston dataset primarily occur in the circular irrigation crop planting areas, but these areas exhibit strong similarity to the surrounding nonirrigated crop planting areas, which requires the model to have a powerful capability to extract change information from the spectral bands of the bi-temporal HSIs. The proposed FreMamba model achieves the best overall visualization results, with lower rates of FPs and FNs. Specifically, in the complex central region of the image, where the changed crop areas are interconnected and exhibit intricate boundaries, the proposed FreMamba model is capable of accurately predicting the actual changes while effectively preserving the fine boundary details of the altered regions. This demonstrates that the proposed FreMamba model possesses strong capabilities in extracting spectral change features and maintaining fine-grained spatial change details. By incorporating the frequency domain-based LSGA and HSGE modules, the method enhances both the spectral and spatial change information, showing its potential for more accurate and robust detection of changes in complex agricultural landscapes.

C. Complexity Analysis

The total parameters and FLOPs of the different methods are listed in Table VII. All the test HSI patch sizes were set to $198 \times 27 \times 27$. Although CDFormer has the fewest parameters, it exhibits the highest FLOPs. The proposed FreMamba model, due to its selective SSM, reduces the redundant computations and, as a result, demonstrates the lowest FLOPs while maintaining a balanced number of parameters.

V. CONCLUSION

In this article, the FreMamba model for HSI-CD has been proposed, which adaptively filters and propagates the critical spectral–spatial information through input-dependent feature selection. First, a Mamba model dual-branch HSI-CD network architecture is designed that separately performs spectral and spatial modeling on the input bi-temporal HSI patches. Second, an LSGA module is integrated into the low-frequency Mamba block to condense the low-frequency features, suppress spectral noise, and mitigate the mixed-pixel problem. Third, the HSGE module is incorporated in the high-frequency Mamba block to extract the high-frequency spatial features, to enhance the spatial geometric information, thereby compensating for the Mamba model's limitations in capturing local spatial details. Finally, in the experiments on three public datasets, FreMamba outperformed the related methods in terms of accurate change information extraction, showing the potential of using the SSM for both low and high-frequency information selection for the HSI-CD task.

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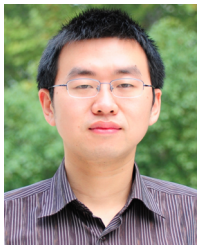
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